
ECE175B Midterm Project Report

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1 Introduction

Recent years, the rise of AI has led to highly sophisticated and intelligent Large Language models - LLMs capable of forming natural and human like language. LLMs have also shown an improvement over traditional machine translations as they are able to incorporate and understand contextual and relational information much better than traditional methods. Gao et al. [2024]

However, LLMs have yet to master the translation of foreign language poems, which is far more difficult than academic or everyday language. Gao [2010-01-01] This project investigates whether model architecture or instruction-tuned decoder-only generation is more effective for low-resource classical Chinese poetry translation under comparable model scale. We compare Qwen2.5-1.5B-Instruct, a small decoder-only instruction-tuned model, with mT5-base, a multilingual encoder-decoder model of similar size ($\approx 580M$ parameters). Both models are fine-tuned using QLoRA on the same training setup and evaluated on the same test set, allowing a clean architectural comparison largely decoupled from scale.

Classical Chinese poetry is linguistically difficult to translate due to its highly compressed, heavily allusive, and poetic features. This limits the resources students and researchers who rely on accurate English translation to study Chinese literature.

2 Related Works

The main problem is that while general-purpose LLMs produce fluent literal translations, they often fail to preserve adequacy, meaning and form. Recent research on LLMs, such as ChatGPT, shows improved accuracy over traditional translation architectures like Google Translate and DeepL Gao et al. [2024]. However, because these large models are trained on overwhelmingly generic internet data, they still fall short of preserving appropriate historical context, concise expression, and poetic elegance. This can be attributed to the lack of existing datasets of high quality Chinese to English translations of poetry. Chen et al. [2025]

To address these limitations, Chen et al. proposed RATs, a fully retrieval-augmented pipeline as well as PoetMT, a benchmark that includes contextual information from external databases to provide richer background for LLM translation Chen et al. [2025]. While RATS effectively improved translation, the pipeline depended on heavy inference making it complex and costly to deploy.

Another relevant line of work is PoetryQwen, which fine-tunes Qwen2.5-14B using LoRA and builds a domain-specific dataset, CCPoetry-49K. Its results show that LoRA can significantly improve performance on classical poetry understanding tasks such as term interpretation, semantic interpretation, and emotional inference. However, its task is mainly poetry appreciation and understanding, not direct Chinese-to-English poetry translation Xie [2025].

3 Method

Addressing the existing limitations, we aim to compare the success of a decoder only model, Qwen2.5-1.5B Team [2024], Yang et al. [2024], with an encoder-decoder model, mT5-base Xue et al. [2021], to evaluate their margin of improvement while limiting computational costs, we implement QLoRA on the smaller Qwen2.5-1.5B model and the mT5-base model.

3.1 QLoRA Implementation

To efficiently update the models by reducing the computational burden of tuning using commercial computers, we utilize Quantized Low-Rank Adaptation (QLoRA). In standard fine-tuning, a pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$ is modified by adding a full-rank matrix $\Delta \mathbf{W}$. At initialization, $\mathbf{A} \sim \mathcal{N}(0, \sigma^2)$ and $\mathbf{B} = \mathbf{0}$, ensuring $\Delta \mathbf{W} = \mathbf{B}\mathbf{A} = \mathbf{0}$ at the start of training so the model preserves its pretrained behavior. LoRA reduces the computation needed by representing $\Delta \mathbf{W}$ with a low-rank decomposition Dettmers et al. [2023].

$$\mathbf{W} = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A} \quad (1)$$

with $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$ and rank $r \ll \min(d, k)$. During training, $\mathbf{W}_0$ is frozen and quantized to a 4-bit Normal Float, while $\mathbf{A}$ and $\mathbf{B}$ contain the trainable parameters. The forward pass becomes,

$$h = \mathbf{W}_0 \mathbf{x} + \frac{\alpha}{r} \mathbf{B}\mathbf{A}\mathbf{x} \quad (2)$$

Where x is the input, α is a scaling hyperparameter and r is a rank hyperparameter

With the implementation of QLoRA, we can expect a 80%-85% reduction in VRAM usage. This allows Qwen2.5-1.5B to be fine-tuned with 3.5-4.5 GB of VRAM and mT5 to be fine-tuned with 1.5-2 GB of VRAM.

3.2 Model Architectures

1. **Qwen2.5-1.5B-Instruct (Decoder-Only):** Our first method uses a lightweight decoder-only model. We utilize the Instruct variant over the base model. The base model acts as a "continuation" machine which are better suited for interpretation within the same language. The Instruct model is already trained with the understanding of "translation" and "English", which forms the foundation of our Chinese poetry translation. By applying QLoRA, we can perform stylistic fine-tuning of the literal translation to elegant English verse.
2. **mT5-base (Encoder-Decoder):** Our second method uses the mT5-base model. The encoder-decoder architecture is inherently more translation-oriented. By implementing QLoRA into the encoder, we fine tune the understanding of cultural references and imagery specific to Chinese poetry. The decoder adapted with QLoRA serves to fine-tune stylistic generation of English poetry.

This comparison tests if the larger Qwen2.5-1.5B model outweighs the translation-centric architecture of the smaller mT5-base.

3.3 Adapter Placement

For poetry translation where the poetic style is a priority. We target the MLP (Multi-Layer Perceptron) where the linguistic "knowledge" of how English sounds can be best modified.

Qwen2.5-1.5B: The Qwen2.5 model uses a decoder-only architecture. We target the attention layers and the MLP layers in this model.

The attention layers $q_{proj}, k_{proj}, v_{proj}, o_{proj}$ ¹ dictate where the model looks to map the syntax of classical Chinese to the structural flow of English poetry.

We then target the MLP layers $gate_{proj}, up_{proj}, down_{proj}$ which transform rigid English to poetic synonyms.

mT5-base: The mT5 model uses an encoder-decoder architecture based on the T5 Raffel et al. [2023] architecture which includes three distinct areas of focus: Self-Attention, Cross-Attention, and the Feed-Forward networks.

¹layer are as described in the source code implementations found on Hugging Face transformers library.

In the Self Attention network, we adapt the q, k, v, o^2 layers to help the model better understand the historical context of the Chinese input. While in the decoder, it maintain the meter and rhythm of the English output.

In the Cross-Attention network, we adapt the q, k, v, o layers to help tie the English decoder to the Chinese encoder’s representation.

In the Feed-Forward network, we adapt the wi_0, wi_1, wo layers similar to Qwen2.5’s $\text{gate}_{proj}, \text{up}_{proj}, \text{down}_{proj}$ to further specialize the model with poetic vocabulary.

While early QLoRA research targeted specific layers to save memory, recent findings have shown that targeting all linear layers yields performances much closer to full fine-tuning. By using QLoRA to save on memory usage, we can keep costs low while targetting all linear layers to optimize the base models thoroughly. Dettmers et al. [2023]

3.4 Training Objective

The training objective is to minimize the cross-entropy loss between the model generated tokens and expert translations provided in the dataset Dettmers et al. [2023]. With a quantized 4-bit normal float weight matrix W^{NF4} , the forward pass is,

$$\mathbf{Y} = (\mathbf{W}^{NF4} \cdot c_1 \cdot c_2 + \mathbf{L}_A \mathbf{L}_B) \mathbf{X} \quad (3)$$

where c_1, c_2 are quantization constants, and L_A, L_B are low-rank adapter matrices used to train.

4 Theoretical Analysis

4.1 Why Low-Rank Adaptation Works

Hu et al. Hu et al. [2021] hypothesise that the weight updates required for task-specific fine-tuning have a *low intrinsic rank* — that adaptation lies in a low-dimensional subspace of the full weight space. For classical poetry translation specifically, the stylistic shift from generic prose to poetic English verse may be even more compact than general-domain adaptation, because the target distribution is narrow and highly structured. This makes LoRA an especially strong fit: a rank-32 adapter captures the relevant stylistic subspace while leaving the vast majority of the pretrained model’s language knowledge undisturbed.

4.2 Why Instruct over Base

The base Qwen2.5-1.5B model is a continuation model: given a classical Chinese poem, it is liable to continue generating Chinese text rather than producing an English translation. The Instruct variant has been aligned via instruction tuning to follow explicit task prompts, including translation requests, so it already produces English output from Chinese input without bespoke prompt engineering. QLoRA can then specialise this aligned behaviour toward poetic elegance rather than first re-teaching the model what “translate to English” means. This is why we use the Instruct variant for both small and large Qwen2.5 paths.

4.3 Decoder-Only vs. Encoder-Decoder Trade-off

The encoder-decoder architecture provides a structural translation inductive bias absent from decoder-only models: mT5’s encoder builds a dedicated representation of the entire classical Chinese source, and the decoder *cross-attends* to that representation at every generation step. This separation may benefit a low-resource task where the mapping from classical Chinese to English is highly non-compositional. Conversely, Qwen2.5-1.5B-Instruct’s instruction-following alignment may allow it to leverage broader knowledge of English poetic register even without an explicit source-encoding bottleneck. Our experiments test which of these inductive biases is more valuable in practice.

5 Experiment Design

5.1 Hyperparameters

We adopt the following hyperparameters based on the works of Dettmers et al. [2023] to capture the semantic shifts and style in poetry.

²layer are as described in the source code implementations found on Hugging Face transformers library.

Rank ($r = 32$): We choose a higher rank than standard ($r = 8$) to capture the nuances necessary to generate specific stylistic writing.

Alpha ($\alpha = 64$): The scaling factor is set to $2 \times r$ to ensure the learned adapters have a significant impact on the model.

Dropout (0.05): A small dropout of 0.05 was found to perform well on small models in Dettmers et al. [2023] and is applied to reduce overfitting.

Table 1 summarizes the five experiments. Experiments E1 and E2 form the primary size-controlled architectural comparison; all other experiments are ablations or reference runs built on top of that result.

Table 1: Experiment design. E1 vs. E2 is the main comparison.

Exp.	Model	Dataset	Purpose
E1	Qwen2.5-1.5B-Instruct+ QLoRA	PoetMT only	Decoder-only baseline
E2	mT5-base + QLoRA	PoetMT only	Encoder-decoder baseline
E3	Qwen2.5-14B-Instruct+ QLoRA	PoetMT only	Large-model upper bound
E4	Best of E1/E2	PoetMT + CCPM ($\lambda > 0$)	Effect of auxiliary semantic grounding

E1 vs. E2 is the primary comparison — architecture is controlled for scale and fine-tuning method, so observed differences are attributable to architectural choice. **E3** is reported separately as a scale reference and is not used to adjudicate the main hypothesis. **E4** is an ablation run on whichever of E1/E2 achieves the higher BERTScore.

5.2 Evaluation Metrics

We evaluate translation quality using three complementary automatic metrics and a small-scale human evaluation.

BLEU-4 Papineni et al. [2002]: Measures 4-gram overlap with reference translations. BLEU is the standard machine translation benchmark but is a known weak signal for poetry, where paraphrase and stylistic variation are desirable rather than penalised.

ROUGE-L Lin [2004]: Measures longest common subsequence overlap between hypothesis and reference, capturing structural similarity beyond exact n -gram matches.

BERTScore Zhang et al. [2019]: Computes token-level semantic similarity using contextual embeddings. BERTScore is more robust to paraphrase and is particularly suited to poetic translation, where surface-level overlap is low but semantic preservation is critical.

Human evaluation (15 poems): Adequacy, fluency, and poeticness are rated on a 1–5 Likert scale following the framework of Chen et al. [2025]. Poems are sampled to cover Tang, Song, and Yuan dynasties.

All automatic metrics are computed on the PoetMT held-out test-set (≈ 73 poems) only. CCPM samples are never included in translation evaluation because CCPM contains no English translations.

Primary comparison (E1 vs. E2): size-controlled architectural comparison between Qwen2.5-1.5B-Instruct and mT5-base.

Scale reference (E3): Qwen2.5-14B-Instruct results reported alongside but not used to adjudicate the main hypothesis.

Ablations (E4): effect of CCPM auxiliary loss relative to the best model from E1/E2.

6 Novelty and Advantages

The primary novelty of our work is to frame classical Chinese poetry translation as *stylistic cloning* via parameter-efficient fine-tuning, rather than as pure linguistic equivalence. Unlike retrieval-augmented

pipelines such as RATs Chen et al. [2025], which incur retrieval and reasoning cost on every query, our approach concentrates the cost in a one-time training pass and produces a deployable model that runs on a single consumer GPU. We do not discard contextual knowledge: instead, we move it from the retrieval stage into the training stage by encoding modern Chinese paraphrases, and creation background directly into the prompt during fine-tuning.

We make four concrete contributions:

Size-controlled architectural comparison. We compare a decoder-only model (Qwen2.5-1.5B-Instruct) against an encoder-decoder model (mT5-base, $\sim 580\text{M}$ parameters) under an identical QLoRA fine-tuning regime. To our knowledge, no prior work has isolated the architectural variable for Chinese poetry translation at this scale.

Scale reference. We report Qwen2.5-14B-Instruct + QLoRA as a separate scale upper bound, trainable on a single 24GB consumer GPU thanks to 4-bit NF4 quantization. This path is not part of the main architectural comparison; it bounds how much translation quality improves with scale alone.

Adapter placement. We adapt both attention and feed-forward layers, motivated by the hypothesis that poetic style is encoded in MLP “knowledge” sublayers and not solely in attention, and consistent with Dettmers et al.’s finding that adapting all linear layers is necessary to match full fine-tuning performance Dettmers et al. [2023].

6.1 Computational Efficiency

A central advantage of our framework is its accessibility on commodity hardware. We monitor:

VRAM Usage: Peak GPU memory during training and inference. QLoRA’s 4-bit NF4 quantization combined with paged optimizers makes fine-tuning the 14B Qwen variant feasible on a single 24GB GPU; the 1.5B and mT5-base paths fit comfortably below 8GB.

Inference Latency: Average wall-clock time per poem. A faster, smaller fine-tuned model may be preferable for deployment even if its quality is marginally lower.

Trainable Parameter Count: The number of adapter parameters relative to total model parameters, quantifying the efficiency of each QLoRA configuration (typically $< 1\%$ of base parameters).

7 Limitations

Despite the contributions above, our work has several limitations.

Compute budget. Within the time and compute constraints of a quarter-long project, hyperparameter sweeps for each path are limited. The size-controlled comparison (E1 vs. E2) holds the fine-tuning method, dataset, and adapter rank fixed across both models, but we cannot rule out that more aggressive per-model tuning would shift the relative ranking.

Dataset size and coverage. High-quality parallel Chinese-English poetry corpora remain scarce. PoetMT is small relative to general-domain MT datasets and is dominated by Tang *shi*-style poetry, which may cause fine-tuned models to overfit to that register and generalize poorly to Song *ci* or other classical forms.

Evaluation of poetic quality. BLEU and ROUGE-L correlate poorly with human judgments of poeticness, and BERTScore — while more robust to paraphrase — was not designed for verse. Our human evaluation covers 15 poems on a 1–5 Likert scale, which gives directional signal but carries variance from a limited annotator pool and cannot be treated as a definitive ranking.

Scale conflation in the upper bound. Path 3 (Qwen2.5-14B-Instruct) is reported as a scale reference, not as part of the main architectural comparison. A direct E3-vs-E2 comparison conflates scale with architecture and should not be read as an architectural conclusion. The architectural claim rests on E1 (Qwen2.5-1.5B) vs. E2 (mT5-base), where parameter counts are within the same order of magnitude.

Generalizability. Our framework is validated only on classical Chinese poetry. While we expect the methodology to transfer to other highly stylized translation tasks (ancient prose, religious texts, indigenous storytelling), this remains a hypothesis until empirically tested.

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